

Adithya Paramasivam

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Education

University of California, Berkeley

Present - May 2023

BACHELOR OF ARTS, COMPUTER SCIENCE AND ECONOMICS · 3.9/4.0

Berkeley, CA

- Course Work: Data Structures (**Java**), Structure and Interpretation of Computer Programs (**Python, Lisp, SQL**), Probability and Statistics (**R**), Designing Information Devices and Systems (**Python, Jupyter**), Discrete Math, Great Ideas in Computer Architecture (**C, Assembly**), Efficient Algorithms and Intractable Problems, Introduction to Machine Learning, Introduction to Deep Learning
- Organizations: Free Ventures (freeventures.org), Upsilon Pi Epsilon (upe.berkeley.edu)

Work

Mantra Bio

May – July 2022

DATA ENGINEER INTERN

San Francisco, CA

- Migrated database queries in **RShiny** reporting application from BigQuery to **Postgres**, with no reported bugs (coverage via **testthat**).
- Built Python **FastAPI** endpoints and added tables to Postgres schema via **Alembic** migrations to process raw full plate NTA (nanotracking particle analysis) .csv data.

Salesforce

May – August 2021

SOFTWARE ENGINEER INTERN

San Francisco, CA

- Reduced redundancies in IP community list data within the DNAGen service (network device bootstrapping), dynamically loading and exposing the data via the Ethgen service.
- Worked within Networking team, implemented the project in **Golang, gRPC** and the datacenter's network topology.

Harmonize Health, Inc.

May - August 2020

SOFTWARE ENGINEER INTERN

Berkeley, CA

- Independently **rebuilt CI/CD infrastructure**, increasing coverage while **cutting code length by 50% and testing time by 70%** and migrating suite from JavaScript to TypeScript. Test suite constructed with **Typescript, PuppeteerJS, cucumber-tsflow**.
- Worked across the stack (**ReactJS, NestJS**) as Harmonize (backed by Trinity Ventures) scaled **3x**, adding providers and thousands of patients.
- Built frontend components with **React, Material UI, RxJS** and deployed web portal with **Docker** containers.

Rabadan Lab, Columbia U.

June - August 2018

DATA/ML INTERN

New York, NY

- Employed the **neural network NN-align** to identify MHC class II binding core and affinity (contributions acknowledged [here](#)).
- Parsed genomic sequencing and RNA-Seq data using **R and Python (w/ pandas, matplotlib, ggplot2)** to narrow down viable biomarker candidates for diffuse large B-cell lymphoma treatments **by 50%**.

Experience

Flipper (link)

Jan 2023 - Present

LEAD DEVELOPER

Berkeley, CA

- Partial reproduction the **FLIP benchmark (link)** for protein sequence-to-function (namely, **CNNs and pretrained ESM models**).
- Leveraged **PyTorch Lightning** to create modular infra for adding models, tokenizers and datasets.
- Overhauled argparse with **Hydra** to set robust configs, which could also be overridden via command line for hyperparameter search scripts.
- Added preprocessing on GB1 and AAV sequences to filter mutation windows, speeding up training and inference by 10x.

Free Ventures

Nov 2019 - May 2022

CO-MANAGING DIRECTOR

Berkeley, CA

- Berkeley's premiere pro-bono pre-seed accelerator, with **alums raising \$85M+ in funding from top firms like KPCB, Greylock, and more**.
- Advising pre-seed stage startups on product development, strategy, fundraising, and general operations.
- Bringing in new mentors (**founders/partners at Superhuman, BoostVC, Initialized, a16z and more**).
- Revamped external marketing strategy for **2x increase** in both batch and internal team applications.

Skills and Interests

Programming Python, PostgreSQL, R, Java, Node.js, Vanilla JavaScript, HTML/CSS, Golang, Lisp, Bootstrap, React.js, Next.js
Technologies Heroku, Amazon Web Services, PuppeteerJS, Pandas, Flask, Apache Maven, Elastic App Search, Jira, Trello